

MODEL QUESTION PAPER

Programname : Mechanical Engineering

Course code:

Course name: **Computer Integrated Manufacturing**

Time : 3 Hours

Max.Marks : 75

1. Answer all the following questions

(9 x 1 = 9 Marks)

1	CIM is the acronym for _____	M1.01	R
2	_____ is used to measure force in automation	M1.02	R
3	Name an output device in CAD	M2.02	R
4	CAD is the acronym for _____	M2.02	R
5	CAM is the acronym for _____	M3.01	R
6	SCARA is acronym for _____	M3.04	R
7	AS/RS is the acronym for _____	M4.01	R
8	CMM is the acronym for _____	M4.02	R
9	_____ is used to measure RPM in automation	M1.02	R

2. Answer any Eight questions from the following

8 x 3 = 24 Marks)

1	Discuss opportunities for automation in production system	M1.01	U
2	Explain any two automation function	M1.02	U
3	Explain features of wireframe modeling	M2.02	U
4	Explain liquid based RP technique	M2.02	U
5	Define group technology	M3.01	R
6	Discuss about features of CNC	M3.04	U
7	List the features and advantages of non-contact measuring techniques	M4.02	R
8	List the factors for selection of manufacturing system	M4.03	R
9	List common types of actuators in automation	M1.02	R
10	Explain DOF	M3.04	U

3. Answer all questions from the following (6x 7 = 42 Marks)

1	Explain programmable automation	M1.02	U
	OR		
2	Describe USA principle	M1.03	U
3	Explain surface modeling with example	M2.02	U
	OR		
4	Explain solid modeling with example	M2.02	U
5	List the advantages of CNC	M3.04	R

	OR		
6	List applications of robot	M3.04	R
7	Explain fixed routing in material handling with examples	M4.01	U
	OR		
8	Explain factors for selection of manufacturing system	M4.03	U
9	Explain ten strategies for automation	M1.03	U
	OR		
10	Explain fixed automation and its features	M1.02	U
11	Describe about Master Production Schedule (MPS)	M3.03	R
	OR		
12	Explain variant type CAPP	M3.02	U

Blue Print

Mark Distribution

Module	Hours/Module (hi)	Marks/Module ($h/\sum H$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of questions	Marks	No. of questions	Marks	No. of questions	Marks	No. of questions	Marks
1	25	42	3	3	3	9	4	28	10	40
2	14	24	2	2	2	6	2	14	6	22
3	20	34	2	2	3	9	4	28	9	39
4	14	23	2	2	2	6	2	14	6	22
Total	73	123	9	9	10	30	12	84	31	123

Cognitive Level Distribution

Cognitive Level	Marks	% of Marks
Remembering	42	34
Understanding	81	66
Applying	nil	
Total	123	100